A060 – Time: Compact Direction, Reciprocity, and Fundamental Beat

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.1 Purpose

[STIPULATION] In FFGFT, time is not an external stage but part of the geometry. From the time-mass duality $\tilde{T} \cdot m = 1$ (A020) it follows in natural units that the native time-energy reciprocity $T{\cdot}E = 1$ holds – an *equation*, not an inequality. This document develops four things from this: the *location* of time (inside the state space, not beside it), the *reciprocity* and its limits (smallest time increment, black hole as extremum, no empty state), the *fundamental beat* (the permitted modes are commensurate multiples of a single ξ-fixed increment), and the *arrow of time* (geometric, not primarily thermodynamic). The cosmically large sector (scales beyond particle physics) is not treated here (A010); it requires its own independently stipulated scale – why, is explained below.

.2 The Fork: Time Inside or Beside the State Space

[B] At the start of every quantum-mechanical modelling stands a mostly unstated choice: *Is time a direction inside the state space, or a parameter beside it?* The question is not technical but ontological. Both answers work with the same formalism – self-adjoint operators, spectra, projectors –; the difference lies solely in *where* time sits. And that exactly decides whether physical quantities *fall out* of the structure or must be *attached* to it.

[B] The question reaches back to the roots of quantum theory. Already in 1932 Pauli recognised that a time operator canonically conjugate to a Hamiltonian bounded below cannot exist (Pauli's theorem); this made time as an external parameter seem incontrovertible. Later work (Aharonov–Bohm, Garrison–Wong) showed that under certain conditions – e.g. for non-semi-bounded spectra – time operators can nonetheless be constructed. The dispute remains unsettled and forms the background to the alternative.

.3 The Unrolled View and Its Price

[B] The familiar formulation places time outside the state space: $\mathscr{H} = L^2(\text{configuration space})$ is the space of states *at one instant*, time the parameter t of the unitary evolution $e^{-i\hat{H}t}$. Observables – position, momentum, spin – are operators *on* $\mathscr{H}$; time is not one (Pauli). The consequence is structural: because time stands outside, the state space alone contains no dynamical scales. Masses, couplings, lifetimes are absent from the pure spatial structure – they are introduced via additional structures, in the Standard Model through Yukawa terms with free parameters. The structure supplies the framework, the numbers come from measurement.

[S] This criticism is old: Weyl and Wigner already saw masses group-theoretically as Casimir invariants – structural labels, not attached parameters –; but the origin of the mass scales remained open. The point is not that the unrolled view is wrong, but that it leaves this question unanswered.

[S] Why attaching is a price. Many programmes with geometric ambition follow a common scheme: (1) a discrete carrier (lattice, cut-and-project structure, periodic graph); (2) a self-adjoint operator family on it; (3) spectral projectors extract sectors; (4) these sectors are *subsequently* assigned meaning – mass often via a parametric map with one or two free parameters. Such a procedure can be mathematically fully controlled; what matters is its *location of time*: the carrier is spatial, the sectors are spatial eigenspaces, not time modes – the mass is not contained but attached. And here an argument applies that is self-supporting: a parametric map with free parameters can hit any finite list of values. If it is not fixed in advance and universally, it merely renames known inputs. To deliver numbers it must be coupled to external values – to measurement, and hence to the Standard Model. An example from established physics: graph-like modelling defines quasiparticles on hexagonal lattices with linear dispersion (massless fermions); a mass always requires additional terms (Haldane, Kekulé) with free parameters. The geometry supplies the kinematics, not the mass scale. The *difficulty* of this path is no coincidence but the direct consequence of the unrolled location of time.

.4 The Rolled-Up View: Time as a Compact Dimension

[B] The alternative does not leave time outside and does not eliminate it, but draws it *into* the state space – as a closed, periodic dimension. In place of $L^2(\text{space})$ with external t stands a space in which a compact time direction is part of the geometry; in the corpus this is $\mathscr{H} = L^2(T^4) \otimes \mathbb{C}^3$ (A070), one torus direction carries the time. On a closed dimension, the permitted states are *standing modes with discrete winding numbers*: quantisation is not postulated but geometrically forced. Mass becomes the *eigenvalue of an operator on $\mathscr{H}$ itself* – not attached but structurally contained. The mass ratios are fixed because the permitted modes are fixed.

[B] Concretely, one expands a function on a compact time direction of length T in a Fourier series,

$$\Psi(t, \mathbf{x}) = \sum_{n\in\mathbb{Z}} e^{int/T}\,\psi_n(\mathbf{x}),$$

the discrete frequencies $\omega_n = 2\pi n/T$ are (in natural units) the mass values, generated by the operator $\hat{E} = -i\,\partial_t$ with spectrum $\{2\pi n/T\}$. Mass quantisation is a direct consequence of periodicity – nothing needs to be postulated. The couplings between modes,

$$g_{ijk} \propto \int e^{i(n_i-n_j-n_k)t/T}\,dt = T\,\delta_{n_i,\,n_j+n_k},$$

are exact selection rules for the mode indices – the "Yukawa couplings" appear as conservation rules, not free numbers.

[B] That a compact dimension generates a discrete mode tower with $m_n \propto n/R$ is the Kaluza–Klein mechanism – here for the rolled-up *time* rather than a spatial direction, with mass in the role of the mode index. And Pauli's objection does *not* apply to this view: it holds for an open, downward unbounded time; a compact time behaves like an angular variable, its conjugate spectrum is discrete (as angular momentum to angle), the Pauli argument does not apply. The closed time is the natural seat of a discrete mode structure.

.5 Time-Mass Duality and Native Reciprocity

[B] The concrete realisation is $\tilde{T} \cdot m = 1$: every massive quantity *is* its own time cycle. In natural units ($E = m$) it follows that

$$\boxed{T \cdot E = 1}$$

This is an *equation*, not an inequality – a sharper statement than Heisenberg's $\Delta E \cdot \Delta t \geq 1/2$, and a native one. It replaces an earlier borrowed justification: the absence of a singular beginning was once argued in the corpus via Heisenberg's energy-time uncertainty. This justification is *replaced*, not refuted – the conclusion stands, the derivation becomes native. Three defects of the borrowed version:

- **Inverted implication.** From $\Delta E \geq 1/(2\Delta t)$, for *finite* Δt a *finite* bound follows, not $\Delta E \to \infty$; the latter follows only from $\Delta t \to 0$, i.e. from the instant $t = 0$, not from a finite beginning.
- **Application error.** By Pauli's theorem there is no self-adjoint time operator conjugate to a bounded Hamiltonian; the strict form (Mandelstam–Tamm) defines Δt as the characteristic *change time of an observable*, not an elapsed duration. An "age" is not a valid Δt of this relation.
- **Style violation.** "Proves irrefutably" is a claim strength that the claim ledger does not know; it knows *derived from ξ*, *restricted*, *blocked*.

[B] The overarching point: Heisenberg's relation is not wrong in the QM framework, but *projected* – it belongs to the description level that FFGFT understands as a projection from the deeper structure (A090). An argument resting on it does not leave the projection but circulates within it.

.6 The Smallest Increment

[K] FFGFT's time is the cumulative deficit of the unrolled winding,

$$D(k) = \sum_{n\leq k} 100\,\xi_n \;\to\; \ln\!\Big(1 + \frac{k}{75}\Big), \qquad \xi_{n+1} = \xi_n\,(1 - 100\,\xi_n),$$

with the first step

$$d_1 = 100\,\xi_1 = 100 \cdot \tfrac{4}{30000} = \tfrac{1}{75}.$$

$D(k)$ grows without bound for $k \to \infty$ (logarithmically); the decisive statement is not "D never diverges" but: $D(k)$ is finite at every finite point, and no time increment becomes smaller than d_1. From $T \cdot E = 1$ it follows that $E \to \infty \Leftrightarrow T \to 0$; a state with $T = 0$ does not exist, because time is bounded below by the finite, ξ-fixed increment. Hence E is bounded above – an infinite energy density is *structurally* excluded, not by a statement about measurement precision.

.7 Both Limits: Black Hole as Extremum, Thinnest State

[K] The same chain determines the black hole – and the result is unspectacular: nothing diverges. From $T \cdot E = 1$ with $T \geq \tau_0$ it follows that $E \leq 1/\tau_0$. The extrema are calculated from ξ: with $L_0 = \xi \ell_P$ (A210) the energy-length reciprocity gives

$$E_{\text{max}} = \frac{\hbar c}{L_0} = \frac{E_P}{\xi}, \qquad \tau_0 = \xi t_P \quad \text{(minimal time scale; by } c{=}1 \text{ the same as } L_0 \text{ in time dress, A080)},$$

and the reciprocity closes *exactly*, because ξ cancels:

$$\tau_0 \cdot E_{\text{max}} = (\xi t_P) \cdot \frac{E_P}{\xi} = t_P \cdot E_P = 1.$$

The same cutoff appears in field theory as the natural UV cutoff at E_P/ξ: the geometry provides the cutoff, a subsequent regulator is not needed (A190).

[K] Scales, not sums. τ_0 and E_{max} are scales, not bounds on totals: τ_0 is the minimal time *increment* of the winding, E_{max} the maximum energy *per excitation* (a cutoff that always bounds a single mode, never a sum). The total energy of a composite system may be arbitrarily large – it consists of many cells, not of a denser one; "the T of the Earth" is not a well-posed quantity. The horizon is accordingly the location at which the *local* density reaches the cell cutoff: slowest time $T = \tau_0$ (not $T \to 0$), maximum concentration $E = 1/\tau_0$ per cell, maximum curvature – all finite and well-defined. It is not a curtain before a singularity but the surface on which time takes its minimum and concentration its maximum. In FFGFT there is therefore *nothing to repair*: models with a bounce or a transition shell fix a defect that FFGFT does not produce.

[S] The other end. $T \cdot m = 1$ has no solution at $m = 0$ – FFGFT has no *empty* state, only a thinnest one. The apparent circularity "vacuum is empty, acts only through curvature, curvature presupposes mass" arises from a causal arrow of general relativity ($T_{\mu\nu}$ *generates* curvature). FFGFT does not have this arrow: $T \cdot m = 1$ is not a cause-effect relation but a reciprocity – mass and temporal curvature are two readings of the same structure, neither generates the other. The circularity dissolves. "Empty" would mean "no winding", and the winding *is* what time is.

[S] The asymmetry, honestly named. The two limits are not equally well determined. The *dense* pole is hard and derived: τ_0 and $L_0 = \xi \ell_P$ follow from ξ. The *thin* pole is structurally excluded, but *uncalculated*: that $m = 0$ is impossible follows from the relation; *how small* m can become minimally does not – and for a structural reason. From $T \cdot m = 1$ every system carries its own upper scale $R_{\text{torus}}(m) = \hbar/mc$; a universal maximum size is not sensibly demanded. For systems beyond the particle scale, a separate, independently stipulated upper bound would therefore have to be introduced – this is not a gap but a consequence of the relation, and it lies outside this document (A010).

.8 The Boundary Question

[B] The torus is compact and boundaryless, $\partial T^4 = \varnothing$ (A090). The question "what was at $t = 0$?" presupposes a boundary at which a beginning could sit; a compact torus has none. The question is therefore not wrongly *answered*, but wrongly *posed*: it imports the decompactified projection as ontology. What an unrolled description reads as a singular beginning is the point at which the *projection* ends – not the world.

.9 The Projection Insight

[S] It is natural to arrange systems on a common axis – from maximum to minimum mass concentration – and to find oneself in the middle. But this axis is an *outside view*: it presupposes a standpoint from which all systems appear side by side. No such standpoint exists in FFGFT. $T \cdot m = 1$ holds *per system*; there is no common T against which to plot. More precisely: we *project ourselves into the middle* – the axis is created with the observer as its zero point, and the middle is not a discovery but a stipulation that goes unnoticed because the one stipulating and the one located are the same. The limits appear as "extreme" because they are far from the chosen zero point: *extreme* means *distant*, not *distinguished*.

[S] The intrinsic time is unaffected by this. For an electron it is irrelevant where an outsider places it on the axis: its winding is the same, because $T = 1/m$ and m is the same. What differs is only how another system sees it – a relation between two, not a property of one. The "slowest time" at the horizon too is a statement about comparison, not about experience. The projection is thereby not invalid but a *tool*: without a common axis, nothing could be compared. It is the same move that resolved the question about $t = 0$ – now applied to the scale itself.

.10 The Fundamental Beat

[K] The permitted states are discrete modes, and they stand in fixed relation to one another – their ratios are exact. From this follows a common *fundamental beat*: through $\tilde{T} \cdot m = 1$ every massive quantity carries its own cycle $T = \hbar/mc^2$, but these cycles are not independent clocks but commensurate multiples of the same ξ-fixed increment $d_1 = 1/75$ – *harmonics of a fundamental beat*, not a continuum of free clocks. This is the precise formulation: not a distinguished base unit against which everything would be measured, but a common beat whose multiples are the modes.

.11 The Arrow of Time

[S] Why does time pass in one direction? In FFGFT the time direction is *geometric*: it follows from the orientation of the winding on the compact time direction, not from thermodynamics. The thermodynamic arrow of time is *secondary* – an emergent property of the non-compact spatial components, not a fundamental given. This explains the apparent irreversibility of macroscopic physics without resorting to ad hoc assumptions about initial conditions. And the time direction is *scale-dependent*: at the fundamental level it is geometrically fixed, at the macroscopic level it appears as a statistical gradient. Both levels do not contradict each other – one is the projection of the other.

.12 Granulation and Fundamental Asymmetry

[S] The deeper reason why time "passes" at all is a *fundamental asymmetry* as the principle of motion: the winding is not symmetric, and the deviation – the deficit $100\,\xi_n$ per step – is the motor that generates time. Reality is thereby *granular*: below the fundamental scale there is no continuum but discrete increments; only above certain scales does time *look* continuous, because the granulation $d_1 = 1/75$ vanishes relative to macroscopic times. This is why practical physics correctly computes with continuous time, even though the fundamental structure is discrete: continuity is a good approximation, not a basic property.

.13 Who Has Already Invoked "Time Not External"

[S] The idea of not treating time as an external parameter is not a novelty but a long-pursued line – and it is honest to locate the rolled-up view within it and to demarcate from it. **Relational time:** Page & Wootters let time arise *within* the state space, as entanglement between a clock subsystem and the rest ("evolution without evolution"); Moreva et al. and Marletto & Vedral refined and illustrated this; Gambini & Pullin developed the Montevideo interpretation. **Timeless quantum gravity:** Wheeler–DeWitt ($\hat{H}\Psi = 0$, the "problem of time"); Kuchař and Isham provided overviews; Rovelli (relational QM, "Forget Time", with Connes the thermal-time hypothesis); Barbour (*The End of Time*). **Compact time:** investigated in string theory; Bars developed a "two-time physics". **Modes as structural eigenvalue:** Wigner (mass as Casimir invariant, a label of irreducible Poincaré representations); Kaluza/Klein (one compactified dimension generates a discrete mode tower with quantised charges).

[S] The demarcation is precise: the overwhelming part of this tradition makes time *relational*, *emergent*, or *timeless* – it *removes* time as a fundamental. The rolled-up view does something subtly different: it *includes* time – as a compact geometric dimension carrying modes. The novelty lies not in "time is not external" (that is old and well established) but in the specific route "compact time generates mass modes".

.14 The Absolute Scale: Fixed, Not Free

[B] The mass ratios follow from the mode structure and are exact; the absolute scale is thereby not yet expressed in SI numbers – but it is for this reason not a *free* parameter. The ratio scaffold rests on the one dimensionless $\xi = 4/30000$. Absolute values carry their correction not as a fit but via fixed *bridge constants*: ν in $m = r\xi^p\nu$ (A100), E_0 in $\alpha = \xi E_0^2$ (A130), fixed factors in G. In ratios these cancel ($m_\mu/m_e = 206.768$, correction-free). $\hbar$ and c are pure SI conversion; a *measured* dimensional anchor in the standard sense does not exist.

[K] The Planck quantities are not a fundamental input but follow from ξ: the chain is $\xi \to \{m_e, \dots\} \to G \to \ell_P$. The *only* exact ξ-power relation is

$$\frac{L_0}{\ell_P} = \xi, \qquad L_0 = \xi\,\ell_P \approx 2.15 \times 10^{-39}\ \text{m};$$

the SI factor ℓ_P must *not* be absorbed into a ξ exponent. The causal direction is therefore reversed: the Planck quantities follow from ξ, not ξ from them. Through $\tilde{T} \cdot m = 1$ the cycle is fixed once the mass is fixed by the ξ-geometry – no gauge modulus remains open.

.15 Causality: Compact Time Is Spectrally, Not Causally, Closed

[B] The obvious concern – does a periodic time generate closed timelike curves? – does not apply. The carrier is a torus ($\pi_1 \neq 0$, stable windings), not a simple circle; the relevant connection is not ontological nonlocality but a *topological* connection on T^4 (A070). The classical CTC problem (Gödel solutions, chronology protection) concerns the *decompactified* Lorentz sector; the compact mass-time direction is *spectrally* closed, not causally.

.16 Verification Scripts

- Native reciprocity and extrema: numerical verification of $T \cdot E = 1$, the exact closure $\tau_0 E_{\text{max}} = t_P E_P = 1$, and $d_1 = 100\xi_1 = 1/75$: `python/A_Serie_Skripte/a060_reziprozitaet.py`
- Fundamental beat: that the mode cycles are commensurate multiples of d_1 (integer ratios), not independent clocks: `python/A_Serie_Skripte/a060_grundtakt.py`

.17 Open Questions

First, the length of the compact time direction is not given as a single SI number. That the ratios are exact and the absolute scale is fixed (not free) via bridge constants is the current state; a measured dimensional anchor in the standard sense does not exist.

Second, the *quantum equivalence* of the unrolled and rolled-up views is known, but its methodological consequences are only systematised, not conclusively evaluated. That both use the same formalism is established; which reading is to be preferred remains a stipulation with an ontological price.

Third, the thin pole is structurally excluded but uncalculated. For systems beyond the particle scale the relation demands its own independently stipulated upper bound; its development lies outside this document.

.18 Supersedes

This document subsumes: Dok. 010 (unit convention, time-energy duality, universal field equation, Lagrange density, characteristic lengths and energy, scale hierarchy), Dok. 069 (time-energy duality, geometric rest-mass definition), Dok. 078 (granulation, fundamental asymmetry, time structure), Dok. 157 (arrow of time), Dok. 296 (time, openness, invariance; observer and projection section), Dok. 306 (native time-energy reciprocity), and Dok. 307 (time in state space, fundamental beat). Incorporated: R50 (the Heisenberg-based singularity argument from Dok. 025/063 is replaced by the native chain $T \cdot E = 1$ – conclusion unchanged, derivation native) and R54 (no N_0 as base unit; modes are commensurate multiples of a common fundamental beat). *Not* taken over: the cosmically large sector (the static ξ-cosmos and background radiation sections from Dok. 069, the speculative sections from Dok. 296 §5, R46) and the scale notation $E_H = E_0\xi^{41/4}$ (calibrated to an externally measured quantity, not a derivation – P35); this area is not treated in the A-series (A010) and requires its own scale. The old $T \cdot m = 1$ singularity proof from Dok. 078 is superseded by the native chain (R50).

.19 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests entirely with the author. Full wording of the rules in A010.